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# Numerical and experimental investigation on the performance of hybrid PV/thermal systems in the north of Iran

Amirhossein Zabihi Sheshpoli [a], Omid Jahanian [a,*], Kamyar Nikzadfar [a], Mojtaba Aghajani Delavar [b]

[a] Faculty of Mechanical Engineering, Babol Noshirvani University of Technology, Babol, Iran
[b] Faculty of Science and Technology, Athabasca University, Alberta, Canada

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ABSTRACT

Solar energy is greatly recommended as a source of renewable energy in wide range of in-house applications to large power plants. Photovoltaic (PV) panels utilized for absorbing solar energy have some limitations: low efficiency, large occupied space and high dependency on environmental conditions. Panel surface temperature is one of the main conditions affecting PV panels. High temperatures would lead to lower panel efficiency. Therefore, researchers have proposed hybrid PV/thermal systems where maintaining panel temperature for higher efficiency is participated in heat production. This benefit made it attractive for household consumption. In this paper, a numerical method is deployed to find an appropriate cooling tube configuration for a specified panel. The selected configuration was prepared for experimental studies on a hybrid PV/thermal system in the northern part of Iran. As this region receives lower irradiation compared to southern parts, People are not persuaded to use photovoltaic systems, so using hybrid systems can motivate them to use solar energy on a small scale. Results showed that serpentine tube configuration would lead to a 7.3% increase in electrical efficiency compared to the no-cooling state, and also thermal efficiency raised to 48.4%. Total efficiency reached 51.76% at the highest performance. Therefore this design provides a possible platform for increasing PV/T systems performance.

## 1. Introduction

For a long time, humans have used all sorts of energy in their daily routines to satisfy their needs without the perception of energy. Today, humans use energy in a variety of fields, such as transportation, heating, cooling, and cooking. Due to the environmental problems caused by the use of fossil fuels as well as the limited resources available to use, countries have been forced to rely on other sources that are not polluting on one side and inexhaustible on the other (Kalogirou, 2004). Renewable energies include wind, water, solar, geothermal, and so on. Among renewable energies, solar energy is more attractive because it is available in more places that other renewable energy sources, besides it is usually simple to use (such as in PV units, flat plate domestic solar water heater). It can be used in small scales like home and large scales as power plants. Furthermore, it can be used in various forms, such as solar water heater, solar chimney, solar air heater, solar desalination, and photovoltaic systems (Olfian et al.,; Smyth, 2019; Hosseinzadeh et al., 2020; Zuhur et al., 2019; Farshad and Sheikholeslami, 2019).

Photovoltaic systems can be a right choice for generating electricity for homes and offices. Despite the advantages of being non-polluting and being silent, they have disadvantages such as low efficiency and variable output during a year. The maximum efficiency of photovoltaic panels under standard conditions (temperature 25 °C, radiation intensity 1000 W/m$^2$, and wind speed 1 m/s) is about 20% (Chandrasekar et al., 2013). Therefore, they require a lot of area for installation to produce acceptable energy for practical uses.

Factors such as rising panel temperature, accumulation of dust on its solar radiation absorbing side, and improper installation will significantly reduce the PV units' efficiency. However, increasing the panel temperature is more critical because about 1 °C increase the panel temperature would decline the efficiency by 0.2–0.5 percent. In other words, if the panel temperature rises 10 degrees due to changes in ambient temperature or radiation intensity, the efficiency of the system drops up to 5 percent of the initial value (García and Balenzategui, 2004). Therefore, given the economic importance of efficiency in large-scale systems, it has led researchers to focus on strategies to counter the effects of rising temperatures to prevent efficiency drop.

* Corresponding author.
*E-mail address:* Jahanian@nit.ac.ir (O. Jahanian).

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**Nomenclature**

| | |
|---|---|
| A | PV Panel area ($m^2$) |
| $C_p$ | Specific heat capacity in constant pressure ($kJ.kg^{-1}.K^{-1}$) |
| $d$ | Tube inner diameter (m) |
| DST | Daylight Saving Time (Hour) |
| f | Friction Factor |
| FF | Fill Factor |
| G | Radiation intensity)$W.m^{-2}$) |
| $h$ | Convective heat transfer coefficient ($W.m^{-2}.k^{-1}$) |
| I | Current (A) |
| $\dot{m}$ | Mass flow rates (kg/s) |
| $Nu$ | Nusselt Number |
| P | Power (W) |
| Pr | Prandtl Number |
| PV/T | Photovoltaic-Thermal |
| $Q$ | Total heat transfer rate (kJ) |
| $Re$ | Reynolds Number |
| $T$ | Temperature ($^oC$, K) |
| $u$ | Velocity ($m.s^{-1}$) |
| V | Voltage (V) |
| $\Delta P$ | Pressure drop (Pa) |
| $\Delta T$ | Fluid inlet and outlet temperature difference)$^oC$, K) |

*Greek symbols*

| | |
|---|---|
| $\eta$ | Thermal and electrical performance factor |
| $\lambda$ | Thermal conductivity ($W.m^{-1}.K^{-1}$) |
| $\mu$ | Fluid viscosity ($kg.m^{-1}.s^{-1}$) |
| $\rho$ | Density ($kg.m^{-3}$) |

*Subscripts*

| | |
|---|---|
| *ave* | Average |
| *b* | Bulk |
| *el* | electrical |
| *Mp* | Maximum power |
| *Oc* | Open-circuit |
| *Sc* | Short-circuit |
| *th* | thermal |
| *w* | Wall |

To avoid the decrease in efficiency due to the increase in panel surface temperature, the researchers have proposed Hybrid Photovoltaic Thermal (PV/T) systems to reduce the photovoltaic panel temperature on the one hand and to use the heat extracted on the other.

To solve the problem of increasing the temperature of the photovoltaic panels, the researchers have investigated the cooling process using air, water, nanofluids, and phase changing materials and also methods such as increasing the heat transfer area by designing different absorbers, and placing a heat pipe and on the PV/T collector (Dupeyrat et al., 2011; Dupeyrat et al., 2011; Das et al., 2018; Pieper and Klein, 2011; Aste et al., 2012). The photovoltaic panel cooling system is designed and utilized by the forced convection and air operating fluid by Teo et al. (Teo et al., 2012). They have observed in their experimental study that the temperature of the panel increased to 68 °C without cooling, and the electrical efficiency decreased to 8.6%. They also showed that with cooling, it was possible to keep the panel temperature fixed at 38 °C and to increase the efficiency to 12.5%. Ibrahim et al. (Ibrahim and Jin, 2009) experimentally investigated the photovoltaic-thermal hybrid system with a rectangular cross-section converter. They examined different airflow rates and, as expected, noted an increase in electrical and thermal efficiencies in all airflow rates. An experimental investigation of a PV/T system with air operating fluid with the rectangular channel was performed by Khurana (Khurana, 2016). He stated that by selecting 0.07 kg/s airflow rate, the efficiency of the panel would reach 9.6%, knowing in the absence of airflow, the efficiency would be around 9%. An experimental study on the cooling effect of a photovoltaic panel with air was carried out by Bambrook and Sproul (Bambrook and Sproul, 2012). They observed the increasing impact of higher airflow rate on thermal and electrical efficiencies. They reported the thermal efficiency of 28–55% and electrical efficiency of 10.6–12.2% if the airflow rate was 0.02–0.1 kg/s. Mojumder et al. (Mojumder et al., 2016) experimentally investigated the effect of temperature on the performance of a photovoltaic panel. In their research, while using air as a cooling fluid, they have used aluminum fins on the back of the panel as heat transfer enhancers. They studied, number of fins evaluated 0 to 4, the airflow rate of 0.02–0.14 kg/s and the intended irradiation of 200–700 $W/m^2$, the results indicated the highest thermal and electrical efficiency as 56.19% and 13.75%, When four fins were utilized, fluid flow and solar irradiation were 0.14 kg/s and 700 $W/m^2$, respectively.

Experimental study on a PV/T collector using water and air as heat transfer fluid was done by Othman et al. (Othman et al., 2013). They checked some designs: double pass PV/T collector with fins, double pass PV/T collector with fin and compound parabolic concentrator (CPC), PV/T collector with rectangular tunnel absorber, and single pass PV/T collector with V-groove absorber. The results show that the collector with only fins performed poorly than the collector with fins and CPC.

Numerical and experimental studies about a proposed heat exchanger design procedure for photovoltaic panel cooling were carried out by Siddiqui et al. (Siddiqui et al., 2019). They found the heat exchanger is more suitable for cooling with less pump work.

Experimental and numerical investigation of photovoltaic panel cooling was performed using a converged channel to obtain uniform temperature (Baloch et al., 2015). The convergence angle of the panel was one of the parameters. The results also showed a 35.5% improvement in the output power of the cooled panel compared to the non-cooled state. Bahaidarah et al. (Bahaidarah et al., 2013) experimentally investigated the cooling of the back panel in hot areas, and their results have pointed to a 20% decrease in modulus temperatures and a 9% increase in efficiency.

The effect of packing factor and water flow rate in in the overall performance of hybrid PV/T systems in areas that have a low intensity of sunlight and ambient temperature was performed by Dupeyrat et al. (Dupeyrat, et al., 2009). The results show that, with low flow rate the performance of this system has been more appropriate.

Khelifa et al. (Khelifa et al., 2016) numerically/experimentally investigated the use of oscillatory tubes for photovoltaic panel cooling. The primary operating temperature of the photovoltaic panel was 60 °C to 80 °C, and they have reportedly reduced this operating temperature by 15–20 percent due to different flow rates.

Adhya et al. (Adhya et al., 2016) performed the cooling of the back panel of PV systems. In their results, they point to an 18 percent increase in panel output power if the temperature drops from 61.6 °C to 41.2 °C by cooling.

Yerevan et al. (Irwan et al., 2015) experimentally investigated the cooling of a photovoltaic panel by spraying water on its surface in a laboratory environment using a solar simulator. Increasing the panel output power from 9 to 22 percent by reducing its operating temperature from 23 °C to 5 °C was one of the significant results of the research.

The performance of photovoltaic thermal collectors with different absorbers design were investigated by Ibrahim et al. (Ibrahim et al., 2009). Various absorber designs were evaluated by changing some parameters such as radiation intensity, ambient temperature, and water flow rate. Based on these simulations, spiral flow design proved to be the

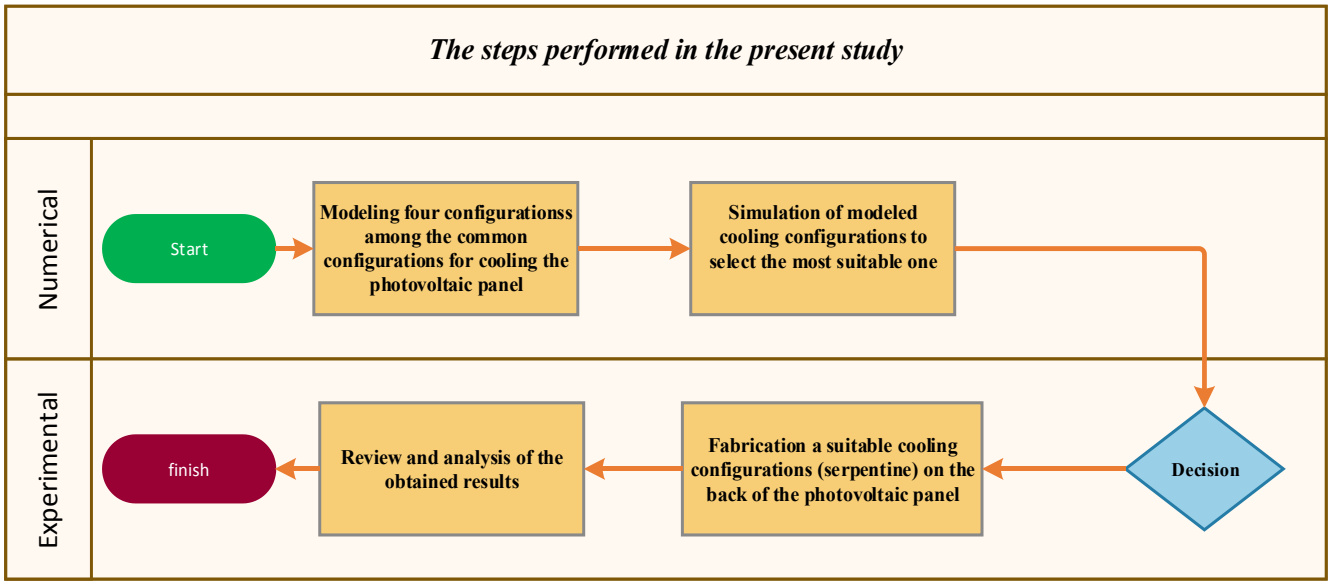



**Fig. 1.** The steps performed in the present study.

best design with the highest thermal efficiency of 50.12% and corresponding cell efficiency of 11.98%.

A new type of covered PVT water system specially desined to be combined with heat pump by Leonforte et al. (Leonforte et al., 2020). In their study, the aluminium roll-bond arrangement was intended for cooling pipes and the absorber was directly connected to the photovoltaic panel. results stated that the system designed with the heat pump can be an alternative to the traditional hybrid PV/T systems.

Also in another similar study, Aste et al. (Aste et al., 2015), Using a a roll-bond flat plate absorber, and embedding it behind a thin film PV technology panel, and using several mathematical equations, investigated the performance of covered PV/T water collectors. In their research results, while pointing out the similarity of the results of experimental tests and dynamic modeling, they have considered the overall efficiency of PV/T system to be higher than the simple PV system.

Another area of interest for researchers in this field has been the use of nanofluids as cooling fluids. Al-Waeli et al. (Al-Waeli et al., 2017) used nanofluid in their research. They used silicon carbide as nanoparticles in their study and noted an improvement of 24.1% in electrical efficiency and 100.19% in thermal efficiency using the nanofluids, as 3% by weight instead of water. Hasan et al. (Hasan et al., 2017) experimentally investigated the nanofluid splash on the back of the photovoltaic panel for cooling down. In their research, silicon carbide, titanium dioxide, and silicon dioxide nanofluids were used. Further research has been done in this area by Li et al. (Li, 2011) as well as Yousefi et al. (Yousefi et al., 2012). In these researches, different types of nanofluids and collectors have been investigated, and their results indicated increasing efficiency of solar systems using nanofluids.

Huang et al. (Huang et al., 2004) numerically investigated the use of phase change materials (PCM) for photovoltaic panel cooling. They realized the dramatic effect of increasing the thermal and electrical efficiencies by incorporating aluminum fins on the back of the panel immersed with PCM. They also stated that the use of more fins would prevent the melting of the phase change materials; thus there are limitations to installing more fins.

Numerical simulations of the use of PCM were carried out by Biwole et al. (Biwole et al., 2013). They delayed the panel surface temperature rise to 40°Celsius for 80 min by using phase change materials, while this time was reduced to 5 min without using the materials.

In another study, numerical and empirical studies were carried out by Aelenei et al. (Aelenei et al., 2014). In their results, they reported the highest thermal and electrical efficiencies of the photovoltaic-thermal system using the phase changed materials to reduce the temperature by 10 and 12%, respectively.

To summarize the case studies, it can be said that each of the different cooling methods has its own advantages and disadvantages. Using water instead of air will improve cooling and adding nanomaterials to the water will increase the heat transfer coefficient. The use of nanofluids instead of water would increase the pump power, and will cause stability problems with the system. Issues such as the cost of phase change materials and the loss of their initial properties after some time are still a hard limitation (Ahmed et al., 2019).

However, in most experimental studies, areas have been selected for the experiment that are rich in terms of the amount of sunlight, but in areas where the radiation is moderate and normal, less research has been done. The northern part of Iran is surrounded by Alborz mountains range and the Caspian Sea, the greatest lake in the world. The mild and wet climate made it a pole of Iran agriculture. Covering only 3.5% of the country area, this crowded area includes about 10% of the population. In contrast with other desert/semi-desert territories of the country, solar irradiation is lower in this part; therefore, there is low interest in utilizing solar-driven instrumentations. Considering the mentioned conditions, persuading people to use solar systems is a bit difficult, so the multiple use of solar systems on the one hand can be a good reason to encourage people to use these systems in the northern region of Iran, and on the other hand can be due to population appropriate in this area, saw the growth of business in the field of solar energy.

In this paper, different cooling tube configurations have been compared numerically to determine appropriate arrangement for a specific hybrid PV/T panel. Then the designed configuration was constructed, and experimental tests were carried out in the described environment. The effects of surface temperature on panel efficiency during the day were comprehensively studied.

## 2. Methodology

A summary of the steps performed in the present study is as shown in Fig. 1.

### 2.1. Numerical modeling

Proper cooling of the photovoltaic panel can result higher performance and longevity. Therefore, choosing the appropriate configuration

**Table 1**
Physical properties for the working fluid and the tube.

| | Material | |
|---|---|---|
| **Property** | *Water* | *Alloy 625* |
| λ (W/mk) | 0.6 | 16.3 |
| ρ (kg/m$^3$) | 998 | 8440 |
| $C_p$ (J/kgK) | 4181 | 505 |
| μ (Pa.s) | $10^{-3}$ | – |

of the tubes in the backside of the panel, which are used for cooling, plays an essential role in the equilibrium temperature distribution of the photovoltaic panel. Numerical simulation is an excellent way to save time and cost compared with experimental efforts. In this study, four different configurations of cooling tubes were created using SolidWorks 17 software and simulated in ANSYS Fluent 19.2 software. Finally, considering parameters such as pressure drop, panel temperature distribution and the optimum thermal index was chosen as the suitable arrangement to be utilized in the experimental study. The designs selected for numerical investigation are chosen based on low cost, and high-performance capability.

The solution has been assumed steady state. In PV/T systems, low fluid velocities are chosen, as it prevents increased pump consumption (Joy et al., 2016; Kalogirou, 2001; Liang et al., 2015; Nualboonrueng et al., 2013). Therefore fluid flow is laminar, and the Reynolds number varies from 200 to 800 by changing inlet fluid velocity. The working fluid is water, and the physical properties can be found in Table 1, as well as properties for Alloy 625, which plates are assumed to be made of.

These assumptions and boundary conditions were applied for simulations:

1. The effect of the PV panel is considered as a metal plate with constant heat flux.
2. The outer wall of the absorber is assumed thermally insulated at the areas in which it does not receive heat flux.
3. Thermal properties are considered constant.
4. The inlet fluid temperature is assumed 300 K. The cold fluid entry in the (a), (c), and (d) configurations is from the top, and in the second configuration is from the left.
5. A constant heat flux equal to 1000 w/m$^2$ is assumed.

### 2.1.1. Governing the equations

Governing equations solved in the numerical model are as below (Schlichting and Gersten, 2016):

Continuity:

$$\frac{\partial(\rho u_j)}{\partial x_j} = 0 \tag{1}$$

Momentum conservation:

$$\frac{(\rho u_i u_j)}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[\mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right] \tag{2}$$

Energy conservation:

$$\frac{(\rho u_i u_j)}{\partial x_i} = \frac{\partial}{\partial x_j}\left(k\frac{\partial T}{\partial x_j}\right) \tag{3}$$

where k is the thermal diffusivity and is defined as:

$$k = \frac{\mu}{pr} \tag{4}$$

The total heat transfer rate is calculated using:

$$Q = mc_p(T_0 - T_i) \tag{5}$$

and the average coefficient of convective heat transfer is obtained from:

$$h = Q/[A(T_w - T_b)] \tag{6}$$

In which $T_w$ is the average wall temperature, and $T_b$ is used for showing bulk temperature. The following equations were also used to compare the performance of each configuration.

Reynolds Number:

$$Re = du\rho/\mu \tag{7}$$

The friction factor:

$$f = 2d\Delta p/(L\rho u^2) \tag{8}$$

Nusselt Number:

$$Nu = hd/\lambda \tag{9}$$

Finally, thermal performance factor is calculated using (Kakac et al., 2020):

$$\eta = \frac{Nu}{Nu_0}/\left(\frac{f}{f_0}\right)^{\frac{1}{3}} \tag{10}$$

The thermal performance factor can be used to analyze the overall thermal enhancement considering simultaneous changes in heat transfer and pressure loss due to the friction.

### 2.1.2. Numerical procedure

In numerical modeling, a rectangular plate is simulated as a panel. The solar radiation above the plate is defined by a constant heat flux, and the cooling copper tubes are considered at the bottom plate. Fig. 2 shows a schematic of the assumed domain.

Also, as shown in Fig. 3, to choose the most suitable configuration for uniform cooling, four configurations a, b, c, and d are selected concerning their fabricability at a reasonable and acceptable cost. For a proper comparison, the length of tubes and the heat exchange area between the flat plate and tube are kept equal for all cases.)The length of configuration (b) is the basis of the design for other configurations.)

The geometrical specifications of the above configurations are listed

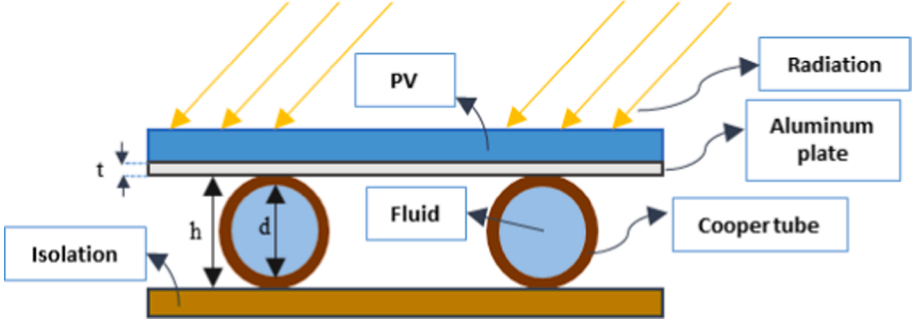



**Fig. 2.** Schematic of the (PV/T) System modeling.

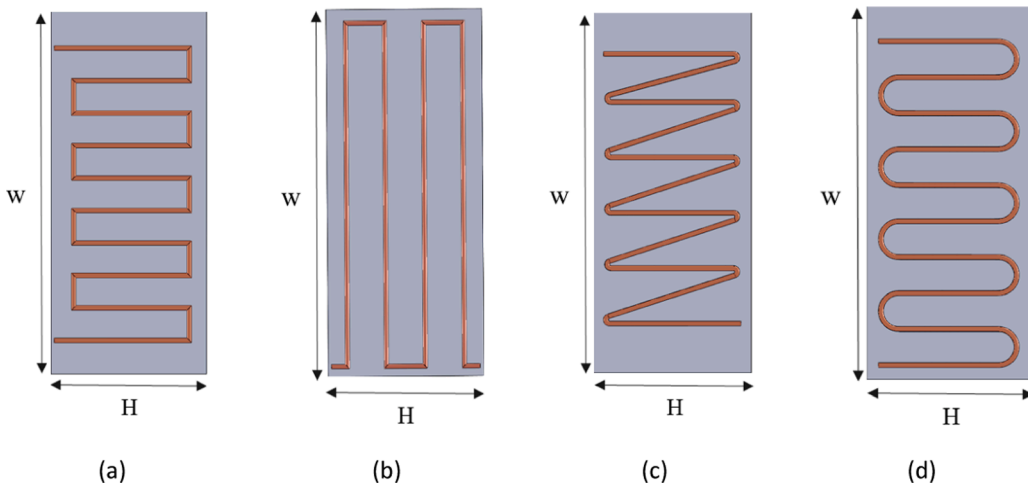



**Fig. 3.** Four configurations selected for cooling tubes.

**Table 2**
Values related geometrical specifications.

| *Tube length(m)* | *H(m)* | *W(m)* | *d(m)* | *R(m)* | *t(m)* |
|---|---|---|---|---|---|
| 6.08 | 1.64 | 0.64 | 0.0178 | 0.0195 | 0.0004 |

in Table 2.

### 2.1.3. Grid generation

Proper mesh generations, would also save time besides enhancing the accuracy of simulation results. For suitable mesh generations, those configurations are selected that are proportional to the geometry of the problem and have a smaller grid size in areas with higher temperature and velocity gradients. Boundary layer mesh is also used near the wall due to the temperature and velocity gradients. The generated grid can be seen in Fig. 4.

Tests for grid independence were performed for various element numbers of 1′716′940 to 4′936′720. The results for the Nusselt number were calculated and tabulated in Table 3. According to Table 3, based on configuration (a), a total of 4′295′817 grid numbers were selected for the study.

## 3. Experimental investigation

As noted above, numerical investigations have been carried out to determine the appropriate cooling arrangement for experimental tests in the north of Iran. For an empirical study on the effects of photovoltaic panel temperature on its performance, a system with precise measuring

**Table 3**
Grid independence tests.

| Number of grid elements | Nu | Deviation in Nu from selected one (%) |
|---|---|---|
| 1,716,940 | 2.944 | −15.8 |
| 3,417,401 | 3.20 | −8.34 |
| 4,295,817 | 3.499 | 0 |
| 4,936,720 | 3.507 | 0.22 |

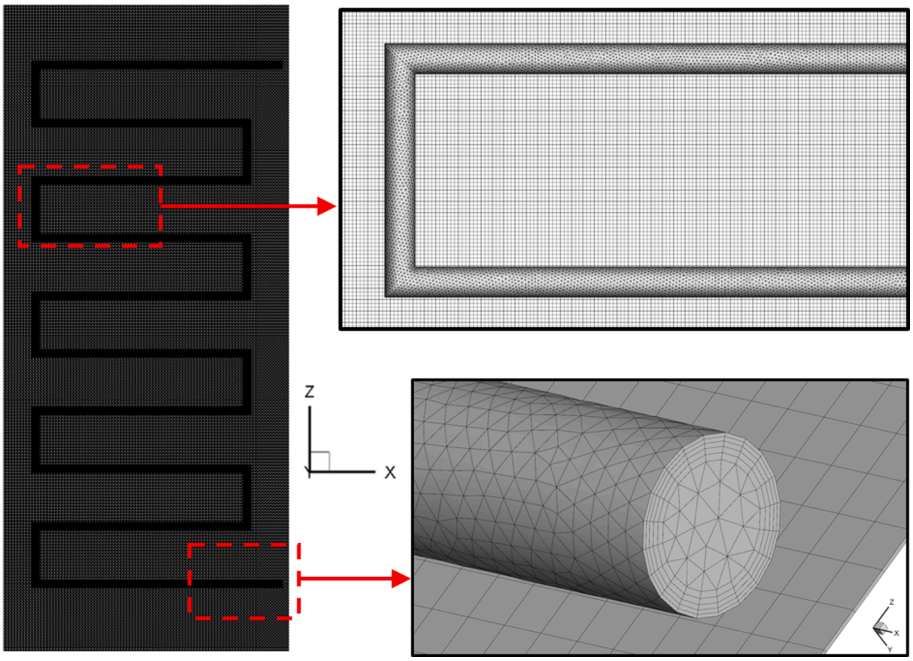



**Fig. 4.** The generated mesh for configuration (a).

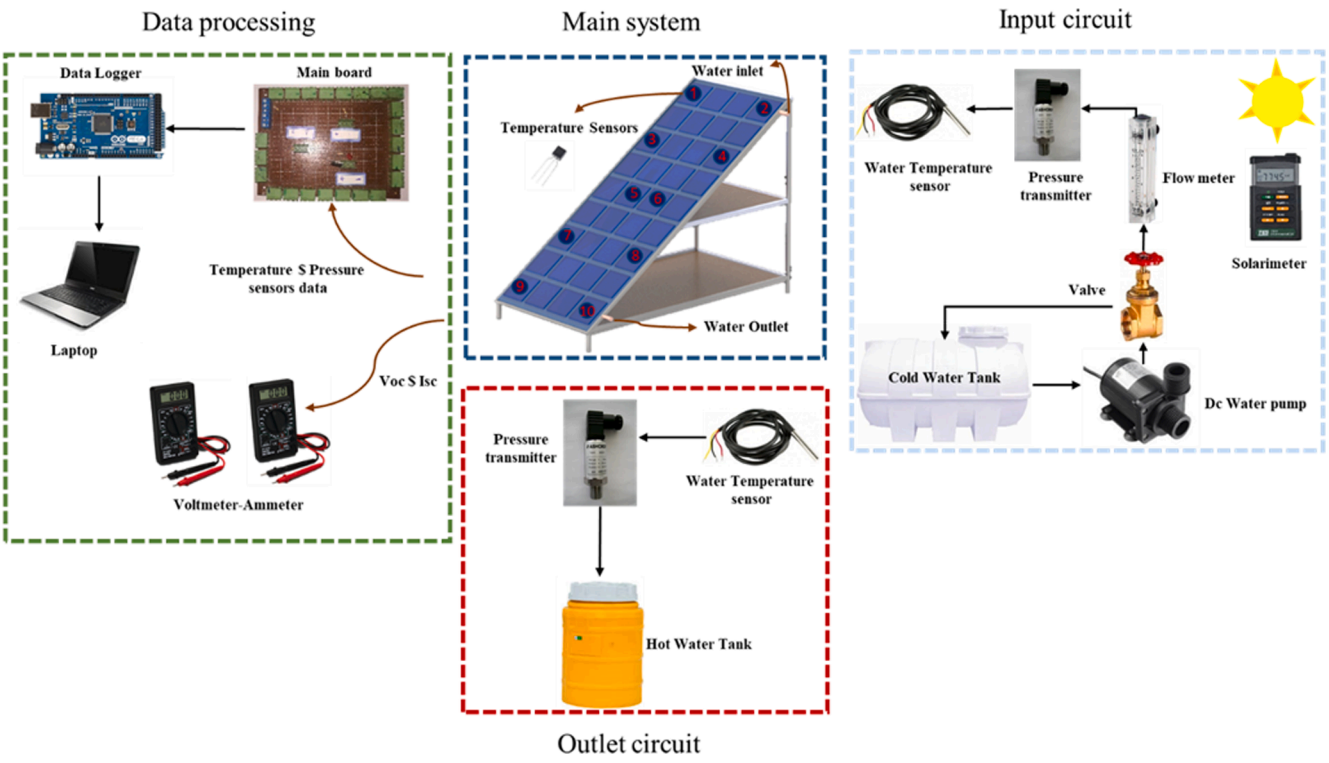



**Fig. 5.** Schematic of the experimental setup.

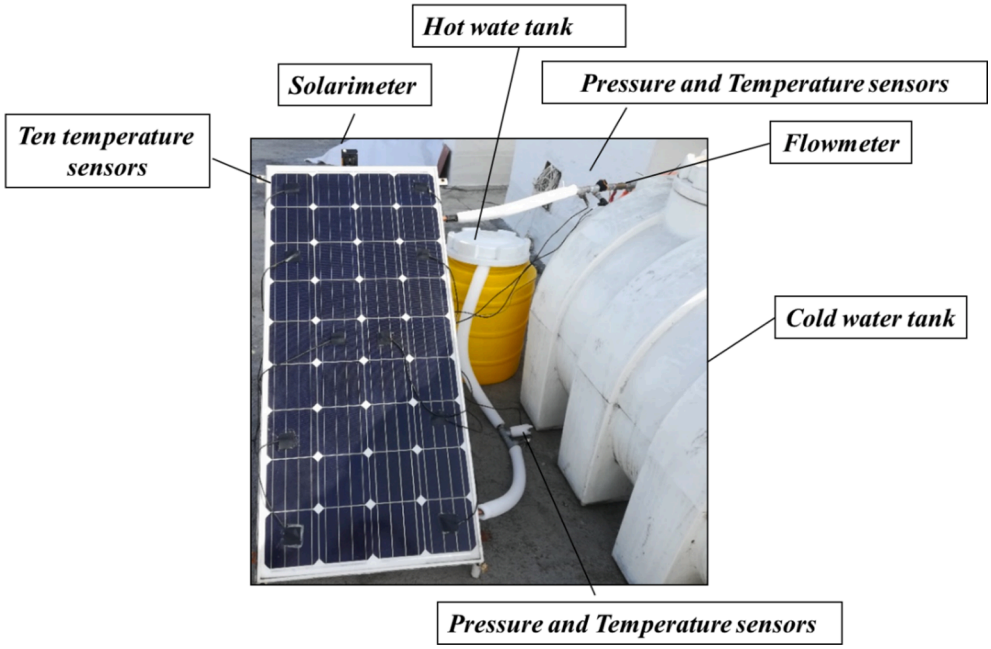



**Fig. 6.** Experimental setup of PV/T system.

equipment is required to achieve the desired results. Experiments were carried out in Babol; a city located in the north of Iran with a geographical coordinate of 36°33′05″N 52°40′44″E during August 2019. The tilt and orientation angles selected 36° and 0° respectively, for proper performance based on geographical location (Duffie and Beckman, 2013).

The cycle was designed to perform investigation of the thermal and electrical performances of a PV/T system, which consists of sensors and measurement and indexing systems. The schematic diagram and actual experimental setup have been shown in Figs. 5 and 6, respectively.

In the experimental setup, a small pump with a maximum power of 5 W, used to take water out of the tank and into the cooling circuit. Although the water flow rate for cooling the photovoltaic panel depends on different parameters such water and ambient temperatures, PV area, and its efficiency, this flow rate is suggested to be 0.001–0.008 kg/s (Joy et al., 2016; Liang et al., 2015; Nualboonrueng et al., 2013). These flow rates were considered to select a pump for the experimental setup. Water has entered the circuit from a reservoir with a volume of 1000 L, and

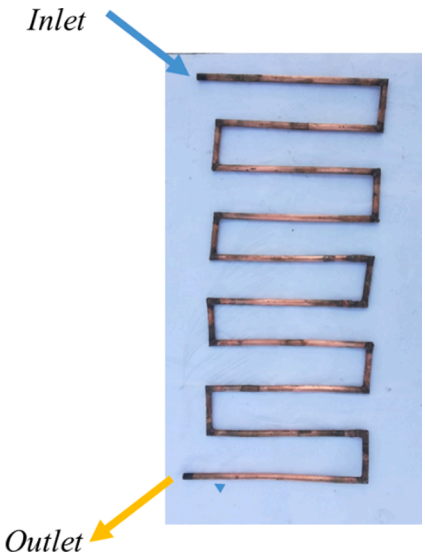



**Fig. 7.** View of the Serpentine cooling tube installed behind the photovoltaic panel.

**Table 4**
Test equipment specification.

| Equipment | Parameter | Experimental uncertainty |
|---|---|---|
| Multi-meter | Voltage | ±0.5% |
| Multi-meter | Current | ±1.8% |
| TES intensity meter | Irradiance | ±5% |
| DS18B20 Temperature Sensor | Temperature | ±0.5 °C |
| Keller Pressure transmitter | pressure | ±0.5% |

**Table 5**
Specifications of the PV module (http://www.yinglisolar.com).

| Brand name | Yingli |
|---|---|
| Model number | YL150C-18b |
| Material | Monocrystalline Silicon |
| Size | 1488 mm × 664 mm |
| Number of cells | 36 |
| Size of a cell | 156 mm × 156 mm |
| Total area of PV cell | 1 $m^2$ |
| Maximum power | 150 W |
| Maximum power voltage (VMp) | 18.52 |
| Maximum power current (IMp) | 8.64 |
| Open-circuit voltage (Voc) | 22.64 V |
| Short-circuit current (Isc) | 8.83 A |
| Standard testing condition (STC) | 1000 W/$m^2$, AM 1.5, 25 °C |
| Efficiency (STC conditions) | 17.8% |
| Fill factor | 0.75 |

after adjusting the flow rate and reading the inlet fluid pressure and temperature, entered the PV/T system. During the test, changes in the panel surface temperature, output temperature, and pressure are recorded. Also, a Digital Multi-meter for measuring short-circuit current, and open-circuit voltage was used. The irradiation is measured by TES 1333 R solarimeter. Ten temperature sensors were used to measure surface temperature, one for ambient temperature and two for water temperature, all of which were DS18b20.

Fig. 7 also shows a view of the Serpentine tube that was installed behind the photovoltaic panel.

The specifications of the equipment used are precisely described in Table 4.

It should be noted that temperature sensors, pressure sensors, and flow meters were calibrated to minimize measurement error. The specifications of the photovoltaic panel used in this study are also consistent with Table 5.

### 3.1. Mathematical formulation

The following relationships have been used to evaluate the performance of the designed PV/T system.

Initially, the Fill Factor (FF) is calculated. This coefficient is calculated according to the specifications of the photovoltaic panel under consideration (Jäger, et al., 2016).

$$FF = \frac{V_{MP} I_{MP}}{V_{oc} I_{Sc}} \tag{11}$$

The electrical output power of the system is calculated according to Eq. (12):

$$P_{max} = V_{oc} \times I_{Sc} \times FF \tag{12}$$

$$\eta_{el} = \frac{P_{max}}{G \times A} = \frac{V_{oc} \times I_{Sc} \times FF}{G \times A} \tag{13}$$

The electrical efficiency of the photovoltaic panel, as well as in the designed PV/T system, is obtained from Eq. (13):

In the above relation, $A$ is the total area of the photovoltaic panel, and $G$ is the intensity of the radiation to the photovoltaic panel at any point in time.

Also, Eq. (14) is used to calculate the thermal efficiency of the PV/T system:

$$\eta_{th} = \frac{\dot{m} \times c_p \times \Delta T}{G \times A} \tag{14}$$

where ΔT is the temperature difference between the inlet and outlet flows.

The following equation is used to calculate the overall efficiency of PV/T system (Wu et al., 2018):

$$\eta_{Total} = \eta_{th} + \eta_{el} \tag{15}$$

As the pump, uses some portion of electrical power for cooling PV, electrical efficiency, Eq. (13), needs to be modified as (Wu et al., 2018):

$$\eta_{el} = \frac{(V_{oc} \times I_{Sc} \times FF) - P_{Pump}}{G \times A} \tag{16}$$

In the above relation, the pump's power is calculated as follows (Kakaç, et al., 2013):

$$P_{Pump} = \frac{m \times \Delta p}{\rho} \tag{17}$$

## 4. Results and discussion

### 4.1. Numerical result

As mentioned before, the first step was to simulate different cooling configurations to achieve the best one.

As shown in Fig. 8, temperature contours are provided for each configuration. Fig. 9 shows the average temperature in terms of Reynolds number. According to these contours and Fig. 9, the configuration (a) has lower average and maximum temperatures than others. For example, the average temperature of the surface in configuration (a) was 20% lower than configuration (d) in the highest Reynolds numbers.

It is observed that by increasing the fluid inlet velocity and consequently increasing the Reynolds number, the cooling rate will increase, and the average plate temperature for all configurations would decrease. But the important point is that there is a slight difference in the average plate temperature between the configurations (a) and (d). According to Fig. 10, this difference originates from a different appearance in the corner of the two configurations. In configuration (a), vortices are created, which make cooling more appropriate. Also, be noted that the

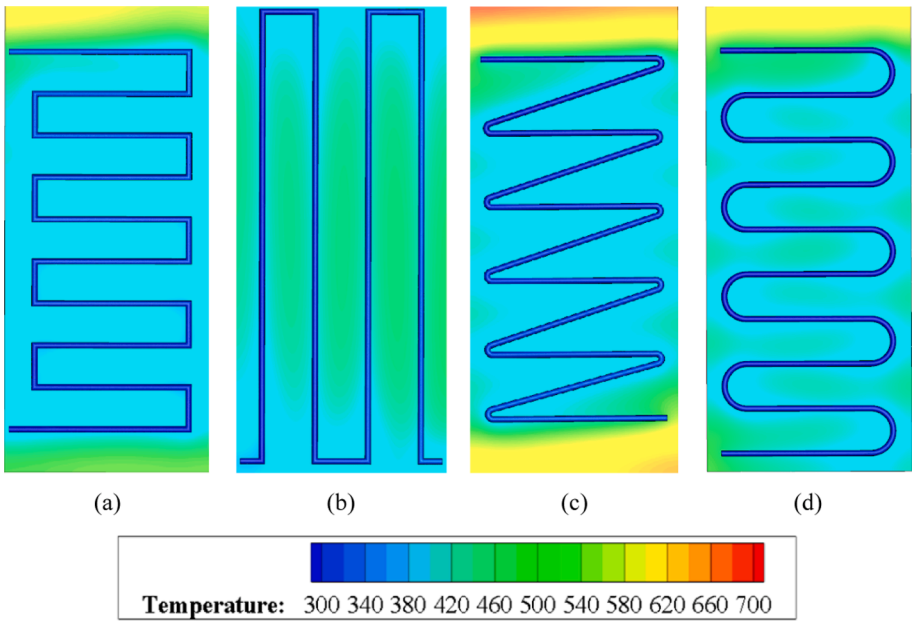



**Fig. 8.** Temperature contour in central horizontal surface of four different tube configurations at Re = 600.

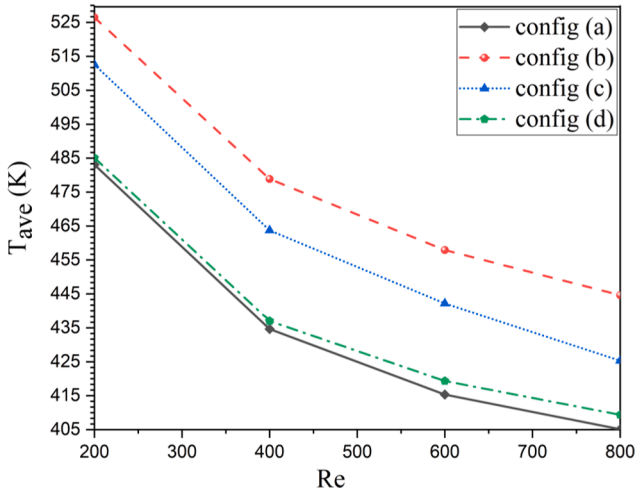



**Fig. 9.** The average temperature of the plate versus Reynolds Number.

average distance between the corners of the tube and the sides of the rectangular surface in the first configuration is less than the fourth configuration. This will cause better cooling of the Aluminum plate's edges.

Nusselt number is considered as another indicator that shows the ratio of convective heat transfer coefficient to the conductive heat transfer coefficient. The Nusselt number curves in terms of Reynolds number are compared for all four tubes configuration (Fig. 11). Configuration (b) has the lowest Nusselt number compared to other configurations, indicating a lower cooling capacity of this configuration than others. This difference appears to be due to the greater length of straight tube in configuration (b) than to other configurations, which results in flow development.

As can be seen in Fig. 12, the first configuration (a) has the highest friction factor compared to the other configurations, and with the increase in Reynolds number, the difference between them increases. The friction factor is used to calculate the pressure drop imposed on the system.

In this study, the thermal performance factor has been utilized for simultaneous considering of the effects of the friction and Nusselt number. As shown in Fig. 13, all configurations were compared

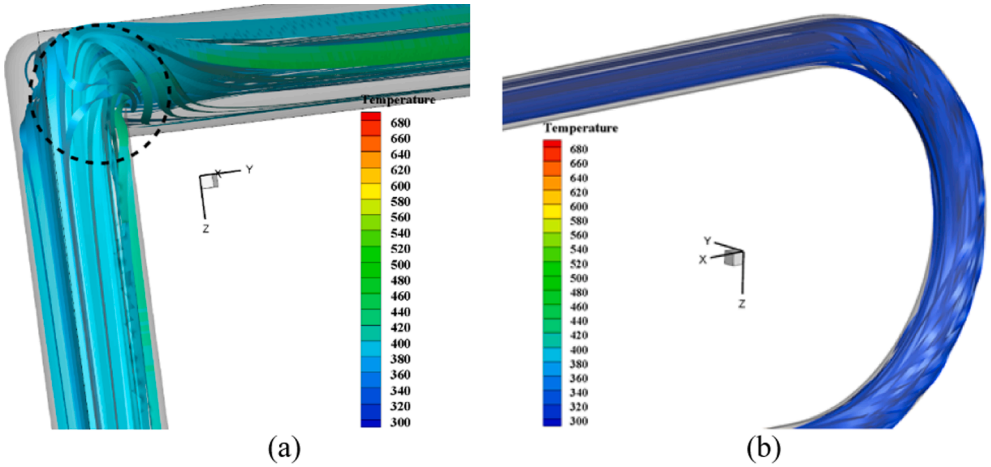



**Fig. 10.** The streamline within the contour of temperature for (a) configuration (a), (b) configuration (d).

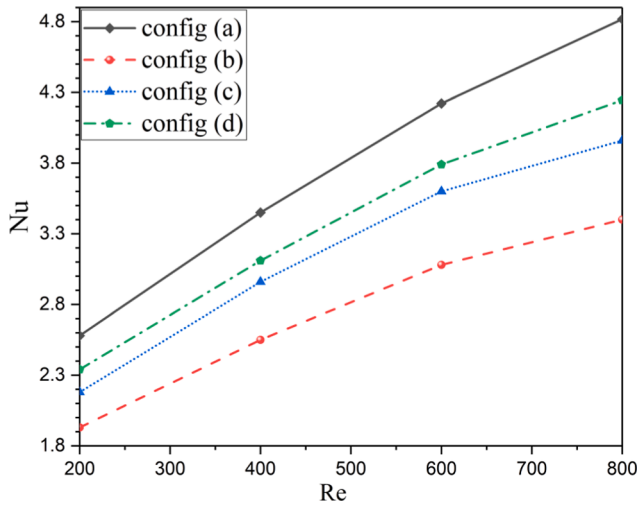



**Fig. 11.** Nusselt Number in term of Reynolds Number.

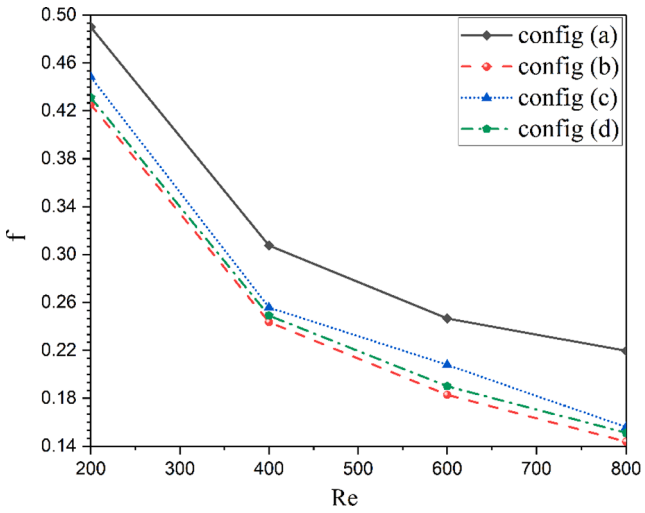



**Fig. 12.** Friction factor in term of Reynolds Number.

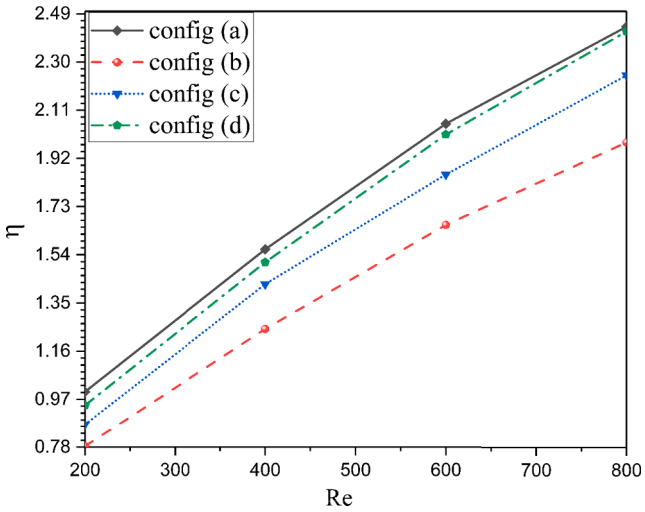



**Fig. 13.** Normalized Thermal performance factor via configuration (a) with lowest Reynolds numbers in term of Reynolds Number.

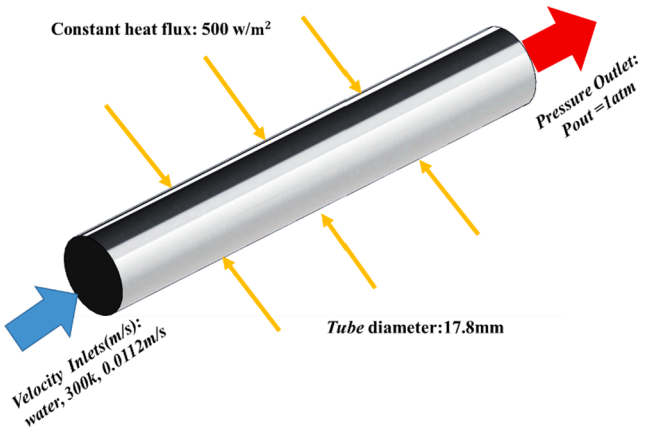



**Fig. 14.** The geometry examined for validation.

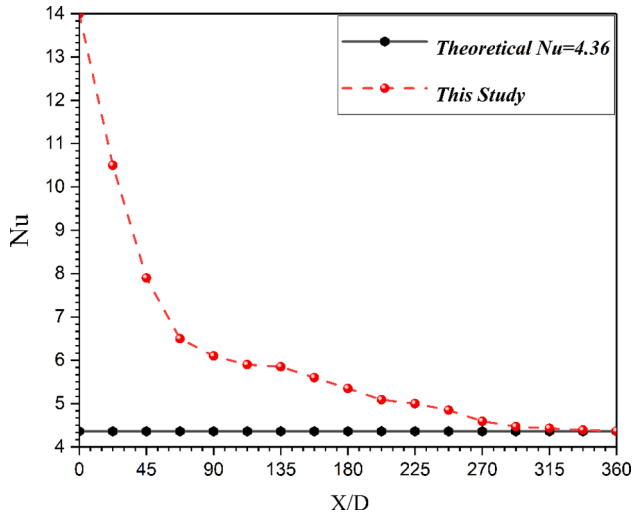



**Fig. 15.** Comparison of theoretical and simulation's result of local Nusselt number for laminar flow in a horizontal tube (Kim et al., 2009).

considering the first configuration with the lowest Reynolds numbers as the base. In the investigated Reynolds number range, using configuration (a) on one side increased the back pressure and on the other side had a better cooling effect. Overall, concerning to the thermal performance factor, configuration (a) in all investigated Reynolds numbers is suggested.

#### 4.1.1. Validation

To check simulation validity, Nusselt numbers for a plain tube in Reynolds number of 200 are investigated. Since the studied flow type is laminar, Nusselt number of a tube exposed to heat flux is fixed at 4.36 for developed flow condition (Kim et al., 2009). Therefore, for validation, the local Nusselt number for the plain tube is investigated. The geometry and results of this study are shown in Figs. 14 and 15.

As the flow develops along the tube, the local Nusselt convergence is proven by the theoretical value, simulation accuracy, and meshing.

### 4.2. Experimental result

After comparing the simulation results in Section 4.1, configuration (a) was selected for the experimental research. On the days of the experimental study, diagrams of changes in irradiation intensity and ambient temperature were recorded and are shown in Fig. 16. It should be noted that the experiments were conducted from 9 am to 5 pm, and the data was stored in intervals of 30 min. Due to outdoor testing, the

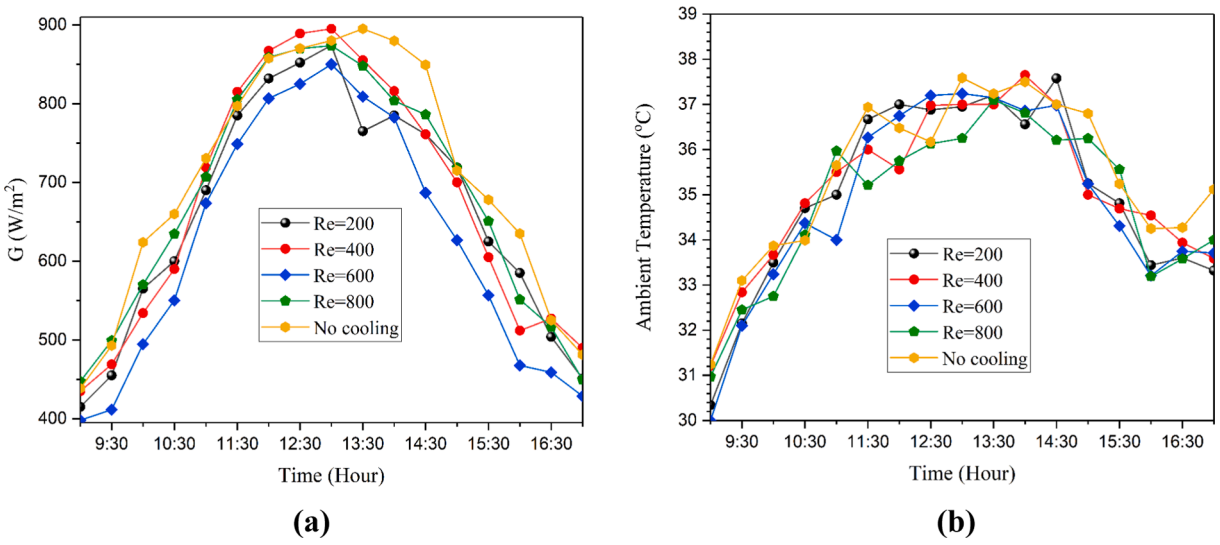



**Fig. 16.** (a) Irradiation intensity, (b) Ambient Temperature in tests carried out for different Reynolds numbers.

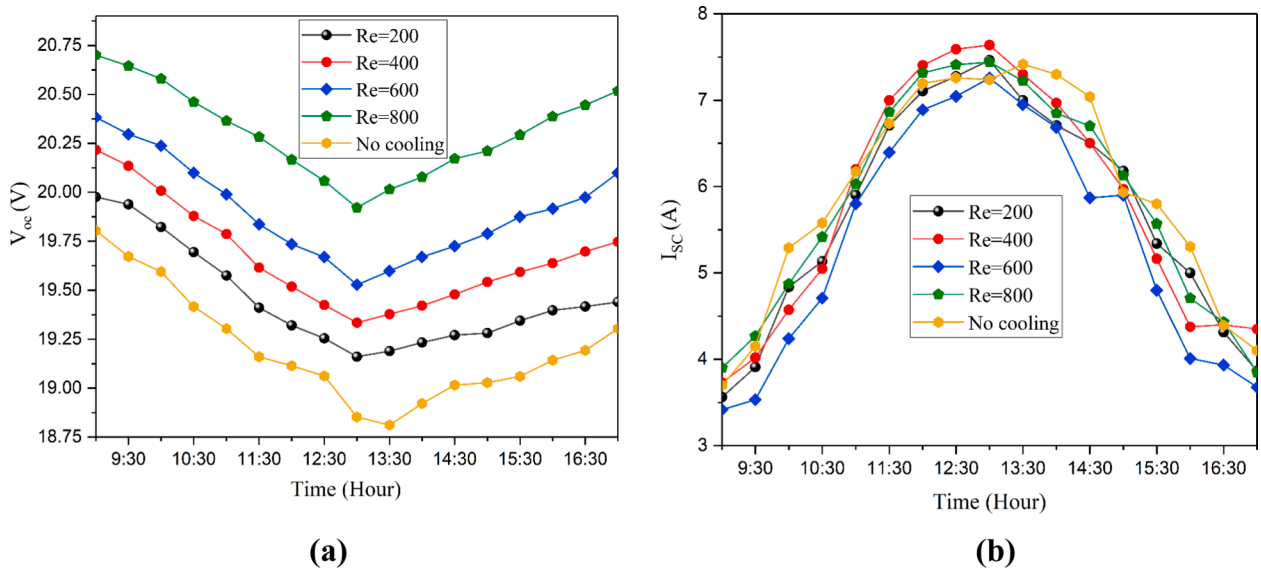



**Fig. 17.** Open-circuit Voltage ($V_{oc}$), (b) Short-circuit current ($I_{sc}$) during the day at local DST.

ambient temperature fluctuations seem reasonable.

Increasing the irradiation intensity reaching the photovoltaic panel on one hand increases the short-circuit current and on the other hand, increases the surface temperature of the photovoltaic panel and consequently reduces the open-circuit voltage. Therefore, as expected, from morning to noon as the irradiation intensity increase, the open-circuit voltage will decrease, as shown in Fig. 17.

The use of the PV/T system reduces the rate of open-circuit voltage reduction compared to simple systems, such as the open-circuit voltage in the PV/T system with a Reynolds number 800 at 13o'clock has improved 4% compared to simple mode.

As can be seen in Fig. 18, with increasing irradiation intensity and ambient temperature, the panel surface temperature also increases, and with decreasing radiation and consequently lowering air temperature, the panel surface temperature would drop. The outlet water temperature of the PV/T system is exactly the same. The water temperature reaches 42 °C in Reynolds 200 at 13 o'clock, while it is about 32 °C at Reynolds 800 at the same time. This difference in temperatures is due to the lower water flow rate and increased heat transfer time. According to $\dot{Q} = \dot{m}c_p\Delta T$, as flow rate decreases in fixed heat transfer rate, temperature difference increases, knowing that inlet temperature is fixed, this would lead to outlet temperature increase.

As shown in Fig. 19, the output power diagram does not have a predictable trend because the irradiation intensity has been different on different test days. The maximum output power of the panel was 110 W during the test days.

Time-varying electrical and thermal efficiencies diagrams are calculated for different modes (Fig. 20). As the fluid flow increases, the surface temperature of the photovoltaic panel decreases, and consequently, the electrical efficiency increases. Thermal efficiency also increases due to the higher heat absorption with increasing fluid flow. According to the results, the electrical efficiency in the PV/T system with a Reynolds number 800 at 13 o'clock has improved by 7.9% compared to simple mode. Also, increasing the Reynolds number from 200 to 800 has increased the average thermal efficiency from 19.76% to 36.92%.

The overall efficiency of the PV/T system is obtained (Fig. 21).

The graph shows that as the mass flow rate of cooling water increases, the total efficiency of the system changes significantly. For example, by changing the Reynolds number from 200 to 800, the averaged total efficiency reached from 34.01% to 51.76%.

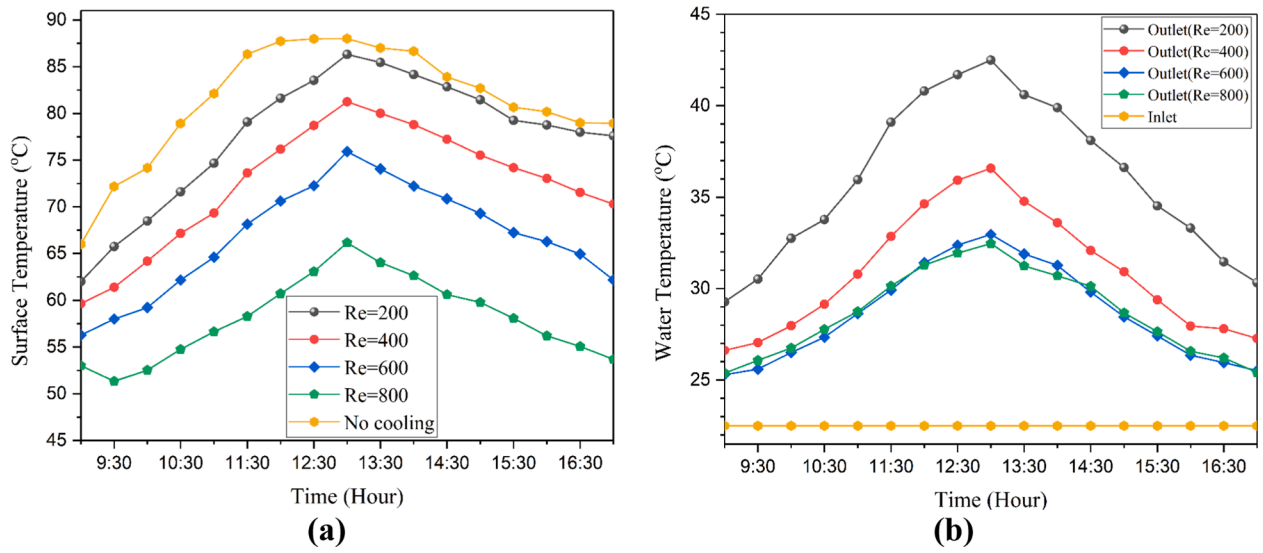



**Fig. 18.** (a) Panel surface temperature, (b) water temperature during the day.

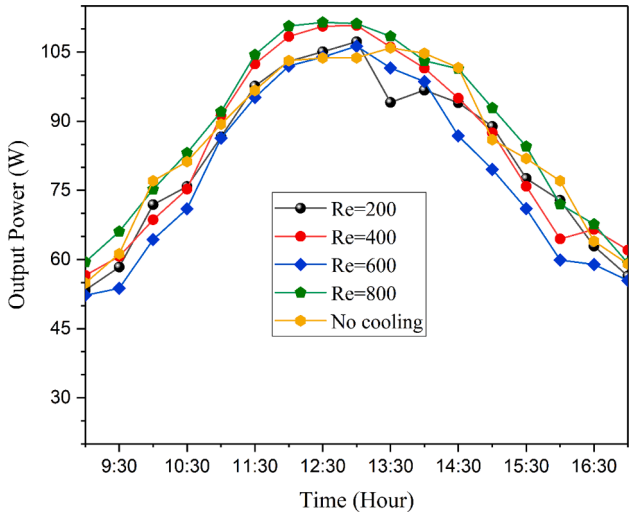



**Fig. 19.** Output power graph during the day at local DST.

It is noteworthy that in many studies in this field, pump consumption has been neglected. However, in this study, we examined and calculated this parameter and observed that by increasing the Reynolds number from 200 to 800, the pressure drop rises up to 7 times. However, due to small flow rates, the pump power consumption is very minimal (less than 0.1 W), so it is ignored.

## 5. Conclusion

In this study, different configurations of hybrid PV/T panel have been studied numerically and experimentally. Cooling of the photovoltaic panel reduces the temperature and consequently increases its service life and efficiency. In previous studies, researchers have noted an increase in electrical efficiency by surface cooling and have also found water as the working fluid to be more suitable than other possible choices. After selecting the appropriate cooling tubes configuration regarding numerical simulations, the application of this system in the northern part of Iran has been investigated. Panel cooling with four different flow rates (proportional to Reynolds number 200, 400, 600, and 800) was performed.

With a hybrid PV/T panel system, electrical efficiency showed an 8% increase at noon. The panel had an average electrical efficiency of 13.83% in the non-cooling state, but with the highest water flow rate, electrical efficiency would rise to 14.84% in the cooled state.

It is noteworthy that by using a PV/T system in addition to increasing the electrical efficiency, the average thermal efficiency during a test day with the highest input flow rate was approximately 37%.

The results show that the average total efficiency during a test day would be 51.76% when the flow rate was at its highest value.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.solener.2020.12.036.

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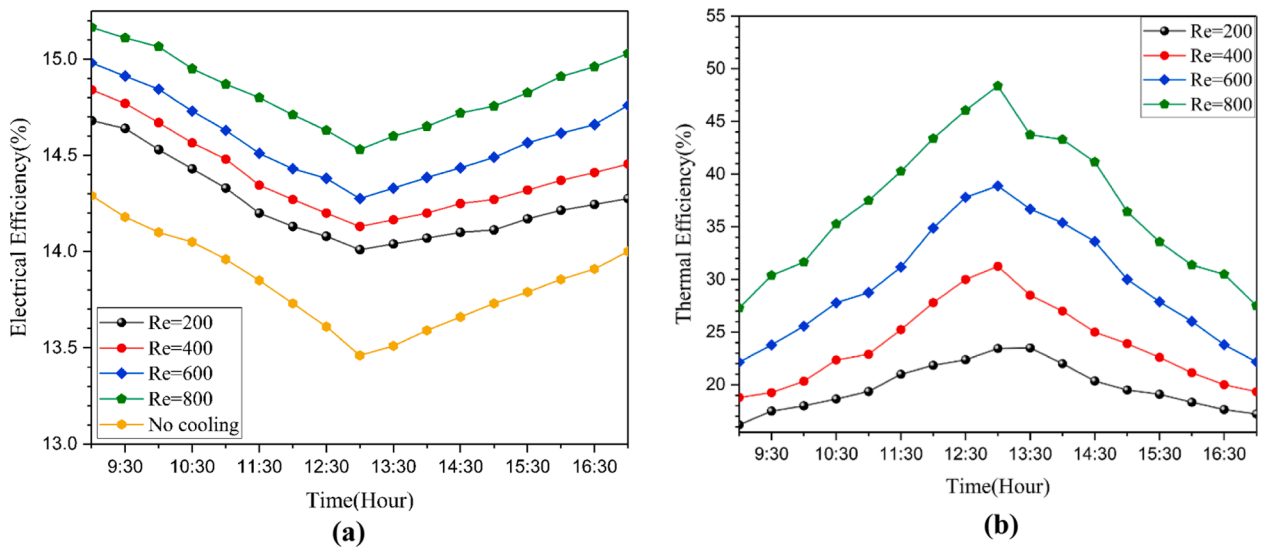



**Fig. 20.** (a) Electrical efficiency, (b) thermal efficiency during the day at local DST.

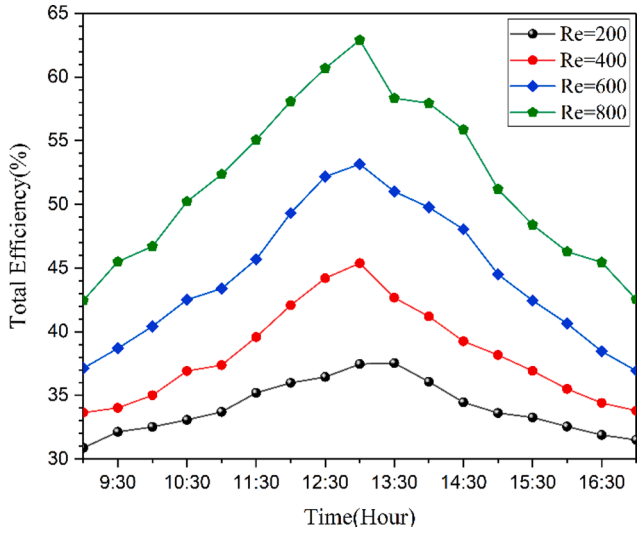



**Fig. 21.** Total efficiency during the day at local DST.


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